Planar Plasmonic film for sensing device applications

The Overview of the Technology:

Surface plasmon resonance (SPR) technology is widely accepted as one of the most powerful approaches for the real time sensing of gas molecules, and real time monitoring of affinity binding of biomolecules at room temperature. The research fraternity and industrialists realised the potential of this technology. However, the technology has been built upon expensive noble metals, gold or silver, for sensing purpose, and hence not economical for cost-effective high volume portable devices. Due to this reason, portable plasmonic sensing devices or devices with disposable sensing chip are not available in market. Other metals are not appropriate for SPR applications since their plasmonic properties are not available in the electromagnetic frequencies of interest. Hence, portable and affordable point of care biosensors require cost-effective plasmonic materials with optimum properties.

C-MET has developed low loss planar plasmonic thin film technology on glass substrates suitable for various sensing applications. These films are cost-effective and can be used in several devices such as gas sensors or biosensors that are currently working with gold or silver. The films developed possess plasmonic sensing effect at par with the noble metals. Portable plasmonic devices can be developed with this planar film, such devices would be highly useful for industries and health sector.

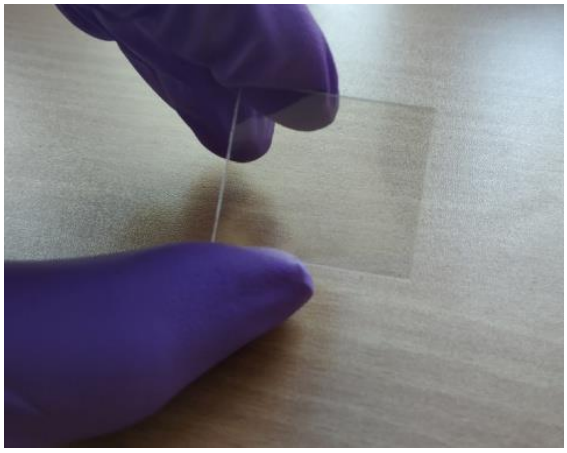
Typical properties of the planar plasmonic film include:

Coating size: 1"x1" on glass substrates

SPR sensing with air as dielectric: 1400±5nm using prism geometry

Tuneability of SPR wavelength (>1400nm) in near infrared can be demonstrated

Glass Substrate: Standard soda lime



1×1 chip

Planar plasmonics film developed at C-MET

The level of technology readiness at C-MET

The technology readiness level is 5

Period of training under the agreement

C-MET will provide in-house training (30 days) at C-MET Thrissur premises and process will be demonstrated for the fabrication of 10 Nos. of 1”x1” size planar plasmonic film on normal glass substrates.

Technology Transfer Fee

Technology for production of Planar Plasmonic Film for Sensing Device Applications: Rs. 4,75,888/- plus applicable taxes.

The ToT package contains the following documents

1. Document containing details of raw material details
2. Details of processes
3. Product specification
4. Procedure for material test
5. Training